TARGETRY AND FEASIBILITY

What would make one of my Ac-225 production plans physically possible?

Right now my equations can rank a schedule that a real target could never survive.

PERSONAL NOTE FOR

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My name is Samuel Ogunnubi, and I am a 12th grader in Maryland. I spent my summer building a physics-guided LSTM for a reduced Ra-226(n,2n)Ra-225 to Ac-225 chain. The neutron pathway I am studying comes from high-energy deuterons on beryllium, which is why your work in radionuclide production, accelerator targetry, radiochemistry, and nuclear data stood out to me.

THE MODEL ONLY SEES ONE LAYER OF THE REAL PROBLEM

PHYSICAL LAYER	CURRENT STATUS	WHY IT MATTERS
Neutron source and transport	Partly represented	A two-group spectrum cannot preserve the full accelerator field.
Target geometry and heat	Not represented	No beam spot, power density, cooling, target damage, or shielding geometry.
Isotope buildup and decay	Represented	A reduced five-nuclide constant-rate chain checked against Radau.
Separation and recovery	Not represented	No chemistry time, recovery fraction, purification loss, or turnaround.

The Berkeley stress test exposed this problem. I pushed the detailed 33 and 40 MeV experimental conditions into my reduced two-group input, even though they were outside the training range, and V3 overpredicted the activity by about 600 times. I kept that failed result instead of retraining on it.

What I can send

- Frozen source, checkpoint, and declared input ranges.
- The reduced reaction chain and solver checks.
- The failed Berkeley reconstruction and schedule-ranking results.

What I need to learn

- The beam, geometry, thermal, cooling, damage, and recovery variables that bound a feasible case.
- A public targetry or radiochemistry benchmark with enough inputs and outputs to reproduce.
- Which constraint would reject an attractive schedule first in real work.

My question: Could you recommend a public benchmark and the smallest set of physical feasibility limits I should add before calling this a production-planning model? Your answer would let me define a realistic domain, reject impossible schedules, and give the surrogate a real job beyond solving a small equation set.